

We thank all reviewers. Most concerns are anticipated in the paper’s objection section (A1–A6); below we point to those pre-emptive answers and add only the new evidence requested.

T_0/α stability (R-isv7). A whole range of α yields near-identical NLL (§A18: three seeds $\alpha \in \{0.34, 0.96, 0.37\}$ within $\Delta\text{NLL} < 0.002$) and near-identical LQ calibration: in a new bootstrap over four backbones the ECE-LQ coefficient of variation is only 3–12% (ResNet 3.0, Swin 4.2, ViT 7.9, ConvNeXt 12.1%). The delivered calibration is thus insensitive to the exact slope—consistent with Prop 1, under which any nonzero slope suffices—so the loose identifiability of α reflects a flat objective, not a fragile calibrator.

Real low-quality data (R-isv7, R-8hpP). QCTS is not evaluated only on synthetic degradation. As pre-empted in A1, we ran EfficientNet-B3 on 174 real-world low-quality dermoscopy images and recovered ECE 0.073 [0.038, 0.125] vs. ITB-LQ’s 0.146 (supp real-LQ table). We are transparent that directly applying QCTS to a real *paired* malignant cohort is future work—no such cohort is publicly available—so the mechanism, not the synthetic pipeline, is the claim.

“Only the weak Std VIB” / structural gain (R-isv7, R-8hpP). Three lines answer this. (i) Std VIB is *by design* the Quality-Fragile repairable prototype ($0 < \text{QCDI} \leq 0.10$) of our taxonomy (§3.2); we deliberately pick a repairable-tier representative to demonstrate the fix, whereas an already quality-aware backbone (ResNet-50) needs no repair (A4, Table 3). Its low AUC is the defining property of the sample, not a flaw. (ii) The mechanism is not confined to Std VIB: using an *external* corruption-severity scalar as $\bar{q}$ (not our IQA module), on both ResNet-50 and ViT-Tiny QCTS makes ρ more negative on 18/18 corruptions, while scalar TS is inert ($|\Delta\rho| < 10^{-4}$, 18/18)—a monotone-invariant clean control. A Wilcoxon signed-rank test over these 18 corruptions confirms significance ($p = 3.8 \times 10^{-6}$). Where there is headroom the gain is sizeable (ViT-Tiny $\rho - 0.138 \rightarrow -0.209$). (iii) On strong backbones (Table 3; ViT-Tiny/ConvNeXt/Swin, AUC-HQ 0.92–0.94) TS induces the QCDI sign-flip ($+0.023/+0.136/+0.020 \rightarrow -0.029/-0.022/-0.021$) and QCTS pulls QCDI back toward zero.

Re-marginalisation baseline (R-8hpP). It is present: the deployment-time re-marginalised baseline is §5.2 Eq.(5), ρ_a (MC-marginalised, $N_{\text{MC}}=20$) = -0.163 vs ρ_b (deterministic- μ) = $+0.241$, a 100% flip; the full table is §A20.

IQA module (R-isv7, R-XHFa). The 5-head module is validated in supp §A2 (mean SRCC 0.895 on held-out synthetic degradations) and the mean-of-5 scalar is justified there as noise suppression; expert-rated validation is a stated limitation. Crucially the reversal-fix also holds under a purely *external* scalar (ImageNet-C severity, above), so QCTS does not hinge on the module.

R-XHFa (factual). QCTS is a two-parameter post-hoc fit on a *single* frozen backbone (§4, L-BFGS for T_0, α), not “calibration across two backbones”—the §5.3 “re-fit” on ResNet, ViT, ConvNeXt and Swin is independent per-backbone refitting to show universality, not a combination. The contribution is conceptual: the QAC/QCDI protocol, the taxonomy, and Prop 1 (scalar TS provably cannot undo a rank reversal, so a continuous slope is the minimal sufficient repair). Reversal holds across $\tau \in [0.40, 0.50]$ (§A10).

R-8hpP (5) / R-6xgi (4). We thank both; the Std VIB scope point is answered via A4 and Table 3, and is a scope—not a soundness—concern.

Honest boundary: the headline magnitude (73% QCDI reduction, halved LQ ECE) is Std-VIB-specific, while the reversal-mechanism *direction* holds across five backbones and 18 corruptions and QCTS never degrades quality-awareness (Table 3).